Lab 3 exo 1 - solution

3.1 - Exercise1:ListeView

1. MainActivity.java

```
package com.example.lab3_1;

import android.app.Activity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import java.util.ArrayList;

public class MainActivity extends Activity {

    ArrayList<Student> studentList;
    StudentAdapter adapter;
    ListView listView;
    EditText inputName, inputDepartment, inputContact;
    Button btnAddStudent;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        inputName = findViewById(R.id.inputName);
        inputDepartment = findViewById(R.id.inputDepartment);
        inputContact = findViewById(R.id.inputContact);
        btnAddStudent = findViewById(R.id.btnAddStudent);
        listView = findViewById(R.id.listView);
        studentList = new ArrayList<>();
        studentList.add(new Student("Alice", "CS", "alice@example.com"));
        studentList.add(new Student("Bob", "CS", "bob@example.com"));
        adapter = new StudentAdapter(this, studentList);
        listView.setAdapter(adapter);

        btnAddStudent.setOnClickListener(v -> {
            String name = inputName.getText().toString().trim();
            String dept = inputDepartment.getText().toString().trim();
            String contact = inputContact.getText().toString().trim();

            if (name.isEmpty() || dept.isEmpty() || contact.isEmpty()) {
                Toast.makeText(this, "Please fill all fields", Toast.LENGTH_SHORT).show();
            } else {
                studentList.add(new Student(name, dept, contact));
                adapter.notifyDataSetChanged();
                inputName.setText("");
                inputDepartment.setText("");
                inputContact.setText("");
            }
        });
```

```
        listView.setOnItemClickListener((parent, view, position, id) -> {
            Student selected = studentList.get(position);
            Intent intent = new Intent(MainActivity.this, ProfileActivity.class);
            intent.putExtra("name", selected.getName());
            intent.putExtra("department", selected.getDepartment());
            intent.putExtra("contact", selected.getContact());
            startActivity(intent);
        });
    }
}
```

2. activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/inputName"
        android:hint="Student Name"
        android:height="50dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <EditText
        android:id="@+id/inputDepartment"
        android:hint="Department"
        android:height="50dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <EditText
        android:id="@+id/inputContact"
        android:hint="Contact Info"
        android:height="50dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <Button
        android:id="@+id/btnAddStudent"
        android:text="Add Student"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp" />

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1" />
</LinearLayout>
```

3. ProfileActivity.java

```
package com.example.lab3_1;
import android.app.Activity;
import android.os.Bundle;
import android.widget.TextView;


public class ProfileActivity extends Activity {
   TextView textName, textDepartment, textContact;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_profile);

      textName = findViewById(R.id.profileName);
      textDepartment = findViewById(R.id.profileDepartment);
      textContact = findViewById(R.id.profileContact);

      String name = getIntent().getStringExtra("name");
      String dept = getIntent().getStringExtra("department");
      String contact = getIntent().getStringExtra("contact");

      textName.setText("Name: " + name);
      textDepartment.setText("Department: " + dept);
      textContact.setText("Contact: " + contact);
   }
}
```

4. activity_profile.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:padding="20dp"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/profileName"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/profileDepartment"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/profileContact"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

5. Student.java

```
package com.example.lab3_1;

public class Student {
    private int id;
    private String name;
    private String department;
    private String contact;
    private int likes;

    public Student(String name, String department, String contact) {
        this.name = name;
        this.department = department;
        this.contact = contact;
        this.likes = 0;
    }

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }
    public String getName() { return name; }
    public String getDepartment() { return department; }
    public String getContact() { return contact; }
    public int getLikes() { return likes; }
    public void setLikes(int likes) { this.likes = likes; }
    public void incrementLikes() {
        likes++;
    }
}
```

6. StudentAdapter.java

```
package com.example.lab3_1;

import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.Button;
import android.widget.TextView;
import java.util.List;

public class StudentAdapter extends BaseAdapter {

    private Context context;
    private List<Student> students;

    public StudentAdapter(Context context, List<Student> students){
        this.context= context;
        this.students= students;
    }
    @Override
    public  int getCount(){return students.size();}
    @Override
    public Object getItem(int position){ return students.get(position);}

    @Override
    public long getItemId(int position) {
        return 0;
    }
    static  class ViewHolder{
        TextView nameText;
        TextView deptText;
        Button likeButton;
    }
    @Override
    public View getView(int position, View convertView, ViewGroup parent){
        ViewHolder holder;
        Student student= students.get(position);
        if(convertView== null){
            convertView= LayoutInflater.from(context).inflate(R.layout.student_item,parent,false);
            holder= new ViewHolder();
            holder.nameText= convertView.findViewById(R.id.textName);
            holder.deptText= convertView.findViewById(R.id.textDepartment);
            holder.likeButton= convertView.findViewById(R.id.btnLike);
            convertView.setTag(holder);
        }else {
            holder= (ViewHolder) convertView.getTag();
        }
```

```
holder.nameText.setText(student.getName());
        holder.deptText.setText(student.getDepartment());
        holder.likeButton.setText("Like (" + student.getLikes() + ")");
        holder.likeButton.setOnClickListener(v -> {
            student.incrementLikes();
            notifyDataSetChanged();
        });
        return convertView;
    }
}
```

7. student_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="horizontal"
    android:padding="10dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:descendantFocusability="blocksDescendants">

    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        android:layout_weight="2"
        android:descendantFocusability="blocksDescendants"

        android:orientation="vertical">

        <TextView
            android:id="@+id/textName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:textSize="18sp"
            android:textStyle="bold" />

        <TextView
            android:id="@+id/textDepartment"
            android:layout_width="match_parent"
            android:layout_height="wrap_content" />

    </LinearLayout>

    <Button
        android:id="@+id/btnLike"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Like (0)" />
</LinearLayout>
```

Note:

the package name depends on your project name.